American BioCarbon® Bagasse-Based Biochar

Technical, Nutritional & Environmental Report

1 Executive Summary

American BioCarbon® (AB) Bagasse-based Biochar is a 100% organic, carbon-rich soil amendment produced from 100% sugarcane bagasse in Louisiana, USA.

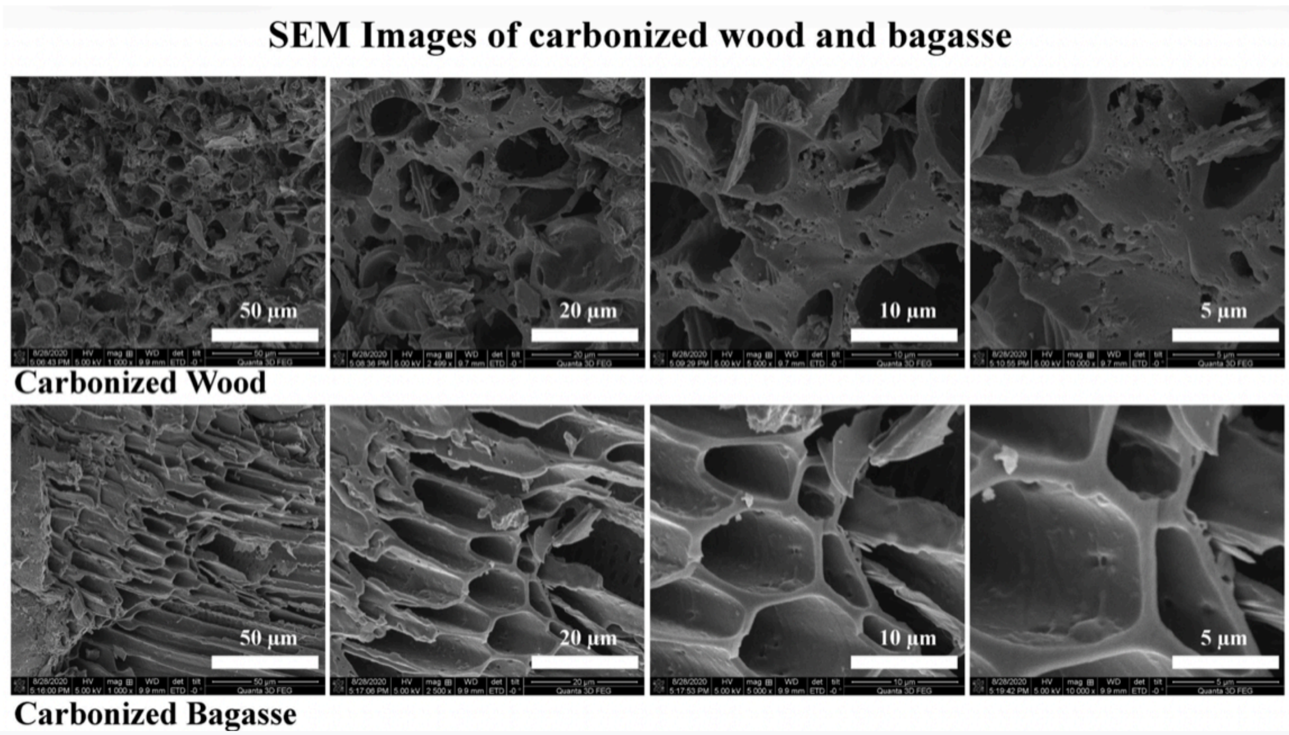
Based on 12 independent IBI laboratory analyses (available on request), the product demonstrates consistent quality, stable carbon structure, measurable inherent nutrients (N-P-K), and extremely low heavy metal concentrations.

AB Bagasse-based Biochar functions simultaneously as:

- A **soil conditioner** (structure, porosity, water retention)
- A **nutrient-retention matrix**
- A **slow-release source of naturally occurring plant nutrients**
- A **long-term soil carbon input**

2 Feedstock & Sustainability

- Feedstock: **Untreated sugarcane bagasse**
- Source: Agricultural by-product of cane sugar manufacturing
- **No trees cut**
- No treated wood, biosolids, or municipal waste
- Renewable, annually replenished biomass



Structural Differences Observed by SEM

Carbonized Wood Biochar

OBSERVED STRUCTURE

- Irregular, fractured pore network
- Predominantly **collapsed vessels and disconnected pores**
- Wide variability in pore size and shape
- More brittle carbon matrix with broken cell walls

IMPLICATIONS

- Lower effective pore connectivity
- Water is held primarily in **macropores**, which drain more readily
- Reduced internal surface area available for nutrient adsorption
- Minimal inherent nutrient contribution (low ash content)

Carbonized Bagasse-Based Biochar

OBSERVED STRUCTURE

- Highly **ordered, honeycomb-like cellular architecture**
- Long, continuous channels aligned with original vascular fiber structure
- Smooth, intact pore walls with high interconnectivity
- Uniform pore dimensions at the 10–50 μm scale

IMPLICATIONS

- Significantly higher **capillary water retention**
- Stronger nutrient adsorption and retention within pore walls
- Enhanced microbial colonization due to protected pore habitats
- Higher mineral ash retention (K, P, Ca, Mg) embedded in the carbon matrix

Functional Comparison (Soil Performance)

Property	Wood Biochar	Bagasse Biochar
Pore geometry	Random, fractured	Ordered, honeycomb
Pore connectivity	Low-moderate	High
Water-holding capacity	**~2× own weight**	**~3–3.5× own weight**
Nutrient retention	Adsorptive only	Adsorptive + inherent nutrients
Ash / mineral content	Low	Moderate-high
Microbial habitat	Limited protection	Excellent protected niches
Performance in sandy soils	Moderate	Excellent
Performance in clay soils	Moderate	Improves porosity & drainage

Why This Matters Agronomically

- **Bagasse biochar** behaves like a **biological sponge**: it stores water and nutrients during wet periods and releases them gradually during drought or nutrient stress.
- The continuous pore channels visible in the SEM images explain why bagasse biochar retains more water, reduces fertilizer leaching, and improves nutrient use efficiency.

Wood biochar, while still beneficial for carbon sequestration, lacks the same pore regularity and nutrient-bearing ash, making it primarily a structural soil conditioner rather than a combined conditioner + nutrient contributor.

Wood-based Biochar -v- Bagasse-based Biochar · Summary Statement

SEM imaging clearly demonstrates that bagasse-derived biochar possesses a uniquely ordered, honeycomb-like pore structure with superior pore connectivity compared to wood-derived biochar. This structural difference directly translates into higher water-holding capacity, improved nutrient retention, enhanced microbial habitat, and superior agronomic performance, particularly in sandy and degraded soils.

3 Biochar Manufacturing Process

- Oxygen-limited thermal carbonization (>500 °C)
- No chemical activators, binders, acids, or bases
- Syngas generated during pyrolysis is captured and oxidized for process heat
- Solid mass reduction ~65–75%, concentrating stable carbon and mineral ash
- All production batches meet IBI H/C molar ratio < 0.7, confirming long-term carbon stability

4 Physical & Chemical Characteristics

Parameter	Typical Value
Organic Carbon	~58–65% (dry basis)
Ash Content	~25–40%
H/C Molar Ratio	0.38–0.61 (≤0.7)
pH	~7.6–9.3
Surface Area	~215–300 m^2/g
Bulk Density	~10–23 lb/ft^3
EC	<0.35 dS/m

These properties support: high water-holding capacity, nutrient adsorption, and microbial habitat.

5 Carbon Stability & Soil Conditioning Function

The honeycomb-like porous structure typical of bagasse-derived biochar:

- Improves soil aggregation and porosity
- Increases plant-available water
- Reduces nutrient leaching
- Provides long-term habitat for beneficial microorganisms

The carbon fraction is highly recalcitrant, supporting decades- to centuries-scale soil carbon retention.

6 Nutritional Value as a Soil Amendment

Unlike many wood-based biochar, AB Bagasse-based Biochar contains measurable, naturally occurring nutrients retained from the original biomass and mineral ash.

Average Nutrient Content (Dry Basis)	
Elemental	
Nitrogen (N)	~6%
Phosphorus (P)	~2%
Potassium (K)	~7%
Secondary nutrients commonly present	
Calcium (Ca)	~0.8–1.5%
Magnesium (Mg)	~1.0–2.0%

FERTILIZER EQUIVALENT

$N - P_2O_5 - K_2O$

N P K

6 – 2 – 7

Important: Nutrients are **inherent, not added**, and released gradually through soil biological processes. AB Biochar contains organic fertilizer, that contribute to soil fertility and nutrient efficiency.

7 Heavy Metal & Trace Element Analysis

Heavy metals in AB Bagasse-based Biochar are far below IBI, EPA Class A, and organic-use concern thresholds.

Average Heavy Metals (mg/kg, dry basis)

Element	Avg	IBI Threshold
Arsenic (As)	~0.55	13
Cadmium (Cd)	~0.31	1.4
Chromium (Cr)	~14.5	93
Cobalt (Co)	~1.5	34
Copper (Cu)	~7.4	143
Lead (Pb)	~2.7	121
Mercury (Hg)	ND	1
Nickel (Ni)	~6.1	47
Selenium (Se)	ND	2
Zinc (Zn)	~105	416

All values are one to two orders of magnitude below limits.

8 Source of Minerals & Trace Elements

Trace elements originate from:

- Naturally occurring Louisiana sugarcane field soils
- Mineral ash retained during carbonization
- Concentration via mass reduction (no addition)

9 Agronomic Performance Summary

AB Bagasse-Based Biochar:

- Improves water-holding capacity (≈3–3.5× its weight)
- Reduces fertilizer runoff and leaching
- Enhances nutrient use efficiency
- Improves performance in sandy and heavy clay soils
- Supports drought resilience

10 Regulatory & Certification Positioning

- Nonsynthetic soil amendment
- Inherent plant nutrients
- Heavy metals below concern thresholds
- IBI certified
- USDA Organic certification (pending)
- OMRI Certified
- Compatible with carbon removal and soil carbon accounting frameworks

11 Intended Use

- Soil amendment and soil conditioner
- Nutrient retention matrix
- Long-term soil carbon input
- Suitable for agriculture, horticulture, land reclamation, and soil remediation

12 Conclusion

> Based on 12 independent IBI laboratory analyses, American BioCarbon® Bagasse-Based biochar is a high-quality, nutrient-bearing biochar that delivers both physical soil improvement and inherent fertility benefits, while meeting the strictest industry and regulatory standards.